Resolution Geometry: The Complete Framework

A Geometry of Observation — Synthesis, Results, and the Case for Implementation

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Abstract

Resolution geometry is a single framework for the geometry of an observation system, organized throughout by one invariant: the coupling between what a measurement resolves and what it does not. This paper assembles the complete machine and states plainly what it offers and what it does not. From one object $\mathcal{G} = (O, g; R, \Sigma)$ we derive, as established mathematics resynthesized around the coupling, the response (Fisher) metric, an estimation theory in which coupling-aware inference beats resolved-only inference by exactly the sector mutual information, a complete classification in which the coupling is provably irreducible, a dynamics with a Mori–Zwanzig memory kernel and an entropy production sourced solely by antisymmetric cross-drift, a reactive response with a derived Lorentzian signature, an identifiability theorem, a global classification as a principal bundle with compact structure group, and a second-order structure that opens the null sector as a second fundamental form. The scientific spine—and the reason to implement—is the bridge between orders: by the Gauss equation the *observable, low-order* curvature is sourced by the *hidden, higher-order* null-sector structure, so cheap first-order measurements encode structure that is otherwise inaccessible. We give an honest impact assessment (quantified estimation gains, hidden-memory detection, model classification, diagnostics, and global observables, each with its caveat), a ledger of the physical claims deliberately retired, and clear boundaries: the framework is a geometry of observation, its global classification is reduced and bounded rather than enumerated, and its formal kinship to physical geometry is resemblance, not a physical theory. Every headline result is numerically verified; the worked machine is exhibited on one example in an appendix.

Contents

1 One object, one invariant

A measurement does not see everything. It resolves some directions of an observation space and is blind to others, and—this is the whole point—the resolved and unresolved parts are statistically and geometrically coupled. Resolution geometry is the systematic study of that coupling. The framework is built from a single object and is carried, by one organizing invariant, from local structure to global classification to second-order geometry.

We are explicit about what this is. It is a *synthesis*: the individual tools (Fisher information, Schur complements, canonical correlations, generalized least squares, Lyapunov and Mori–Zwanzig equations, principal bundles, holonomy, the second fundamental form) are classical, and we credit them as such. The contribution is their organization around the resolved–null coupling as the primary invariant, together with a small number of specific new theorems and one genuinely striking identity. We are equally explicit about what this is not: it is not a physical theory, and the physical claims of earlier drafts have been retired, on the record, in Section 6.

2 The object and the invariant

Definition 2.1 (Resolution geometry)**.** A resolution geometry is $\mathcal{G} = (O, g; R, \Sigma)$: an observation space O with metric g, a resolved sector $R = \mathrm{Im}(J)$ (the range of the response Jacobian), a null sector $N = R^{\perp_g} = \ker(J^*)$, and a covariance Σ. In a g-orthonormal adapted frame,

$$\Sigma = \begin{pmatrix} \Sigma_R & C \\ C^\top & \Sigma_N \end{pmatrix},$$

and the resolved–null coupling $C = P_R \Sigma P_N$ is the primary invariant.

Everything below is a statement about C: how it rigidifies symmetry, how it improves inference, why it is irreducible, how it is generated dynamically, how it sets the reactive signature, how it globalizes, and how it reappears at second order as curvature.

3 The machine, layer by layer

Each layer states its load-bearing result. All are proven and numerically verified in the companion papers; here we assemble them.

3.1 Static: the response metric is derived, and coupling rigidifies symmetry

The Gaussian response (Fisher) metric is not postulated but derived: $\mathcal{I} = J^\top \Sigma^{-1} J = J_R^\top (\Sigma_R - C\Sigma_N^{-1} C^\top)^{-1} J_R$, an inverse Schur complement. The automorphism group is the coupling stabilizer; when $C \neq 0$ it is strictly smaller than the product of sector isometry groups, so coupling rigidifies symmetry.

3.2 Inferential: coupling-aware estimation is cheaper by exactly the information the null carries

Theorem 3.1 (Efficiency = information)**.** *For a calibrated null baseline, coupling-aware generalized least squares dominates resolved-only estimation, and the log-volume efficiency gain equals the sector mutual information:*

$$\tfrac{1}{2}\log\frac{\det V_{\mathrm{res}}}{\det V_{\mathrm{full}}} = I(y_R; y_N) = \tfrac{1}{2}\log\frac{\det\Sigma_R\,\det\Sigma_N}{\det\Sigma}.$$

This is the framework's most striking result: two quantities from different subfields—estimation efficiency and information content—are the same number, exactly. It is also the clearest practical offering (Section 5). Verified: both sides = 0.136775 on the worked example.

3.3 Classification: the coupling is irreducible

Resolution geometries are classified up to admissible isomorphism by a complete invariant (block spectra, whitened-coupling canonical correlations, and the automorphism group); the moduli dimension is $rq{+}r{+}q$. The coupling is *irreducible*: canonical correlations alone are an incomplete invariant for $r, q \geq 2$ (there is an explicit residual the correlations miss). This is a real theorem with a real negative inside it.

3.4 Dynamical: memory and the arrow of time

The coupling is generated by cross-dynamics through an exact Mori–Zwanzig kernel $K(\tau) = A_{RN}e^{A_{NN}\tau}A_{NR}$; the stationary coupling solves a Lyapunov equation. Entropy production is $\sigma_{\mathrm{EP}} = \mathrm{tr}(\nu^\top D^{-1}\nu\Sigma)$ and vanishes iff detailed balance holds; its sole source is the *antisymmetric* cross-drift. A linear–additive model stays Gaussian (a proven ceiling): genuine complexity requires nonlinearity.

3.5 Reactive signature: a derived Lorentzian sign

Semi-win 3.2 (Reactive Lorentzian signature)*.* The reactive part of the response, $\mathrm{Herm}(\chi(\omega))$, is congruent to $K - \omega^2 M$; by Sylvester's law of inertia its signature is the inertia of $K - \omega^2 M$: positive-definite below the fundamental resonance, and exactly one negative eigenvalue in the first window—a Lorentzian $(1, n-1)$ signature—with the dissipative part sign-locked by the second law. The negative sign is *derived* (stiffness minus inertia $\omega^2 M$), not inserted.

This is a semi-win in the exact sense: a genuine, exact result (verified for $n = 2 \ldots 5$; below fundamental $(4, 0)$, first window $(3, 1)$) whose tempting physical reading is deliberately *not* claimed. The signature is a congruence invariant of a response form; it is not spacetime curvature, not general relativity, and frequency-local (Section 7).

3.6 Reconstruction: identifiable up to gauge, but not from Fisher

The geometry is a canonical, gauge-equivariant function of exact observable data (J, Σ, g), identifiable up to admissible gauge, with no non-identifiability witness. But the Fisher metric alone does *not* identify it (it is an inverse Schur complement, blind to Σ_N when $C = 0$): reconstruction requires the full covariance.

3.7 Global: a compact bundle, reduced and bounded

Admissible transitions form a groupoid; a coherent atlas is a principal $\mathrm{Aut}(\mathcal{G})$-bundle. The structure group is a *compact* Lie group (explicitly $\mathrm{O}(r)\times\mathrm{O}(q-r)$ for the canonical coupling), so transport is bounded (the amplification diagnostic is the compactness test). Existence is the Čech cocycle class; flat geometries are classified by the character variety $\mathrm{Hom}(\pi_1(B), \mathrm{Aut}(\mathcal{G}))/\mathrm{conj}$. Not computed, but fully bounded: compact, real-algebraic, of explicit dimension, symplectic on its smooth locus.

3.8 Second-order: the null sector opens as curvature

The Hessian's normal component is the second fundamental form $\mathrm{I\!I} = P_N(\nabla^2 Y)$, valued in the null sector and invisible to first order; the null sector, passive at first order (via C), becomes geometrically active at second order (via $\mathrm{I\!I}$), carrying a normal connection whose normal holonomy is a second global datum. This is the bridge of the next section.

4 The scientific spine: lower curvature is sourced by higher curvature

The reason to implement the framework is that it connects what you can cheaply measure to what you cannot otherwise see, and the connection is an exact classical identity.

Theorem 4.1 (The curvature bridge)**.** *The intrinsic curvature of the observation manifold—* a first-order, observable *quantity, computable from the induced (response) metric—is, by the Gauss equation, sourced by the second fundamental form* $\mathrm{I\!I}$*, a* second-order, null-sector *quantity that first-order observation cannot access directly:*

$$K_{intrinsic} = \frac{\langle \mathrm{I\!I}_{11}, \mathrm{I\!I}_{22} \rangle - \langle \mathrm{I\!I}_{12}, \mathrm{I\!I}_{12} \rangle}{\det g}.$$

Hence measuring the low-order curvature constrains the hidden higher-order null-sector structure.

Verified: on a worked map the intrinsic (metric) curvature and the $\mathrm{I\!I}$-curvature both equal -0.51908; on a validating sphere the $\mathrm{I\!I}$-curvature returns $1/R^2$ exactly.

Corollary 4.2 (Two observable windows on hidden structure)**.** *The framework offers two low-order windows on high-order/global structure: the intrinsic curvature (Theorem 4.1) exposes the second fundamental form (null-sector, second order), and the geometric phase of the reactive time-axis exposes the global holonomy (the character-variety point). In both, a quantity you can observe reports on a quantity you cannot.*

$$\boxed{\begin{array}{c}\text{lower curvature}\\ \text{(intrinsic, observable)}\end{array}} \xleftrightarrow{\text{Gauss equation}} \boxed{\begin{array}{c}\text{higher curvature}\\ (\mathrm{I\!I}\text{, null sector, hidden)}\end{array}}$$

This is the implementation rationale in one line: *the coupling is statistical at first order, geometric at second order, and the two are joined by an exact equation*, so first-order data is a lever on second-order structure.

5 What the framework offers (honest impact)

Each offering is stated with its quantified benefit and its caveat.

1. Cheaper, better estimation. When null-sector observations (references, nuisance channels) are correlated with the resolved signal, coupling-aware estimation recovers precision equal to the sector mutual information $I(y_R; y_N)$ (Theorem 3.1)—an exact, computable gain. *Caveat:* the gain is real only for a *calibrated* null baseline; a miscalibrated null can produce an illusory gain (a documented dichotomy). This is the framework's most directly useful result.

2. Hidden-memory / latent-structure detection. Two systems identical at first order but with different hidden null-sector memory produce different observable couplings (via the Mori–Zwanzig kernel), so the framework detects latent dynamical structure from observable geometry. *Caveat:* requires stationarity and access to the coupling estimate.

3. Model classification and equivalence. The complete invariant gives a canonical form and decides when two observation systems are the same geometry (useful for deduplication and equivalence testing). *Caveat:* classification is up to admissible gauge and fixed similarity type.

4. Model diagnostics. For multi-chart models the framework localizes inconsistencies (a bad transition corrupts only adjacent plaquettes), repairs them constructively (nearest admissible transition by polar projection), and flags ill-posed transport (unbounded iteration = leaving the compact structure group). *Caveat:* global repair is an open optimization.

5. Global / topological observables. The geometric phase of the reactive time-axis around base loops measures the global holonomy, giving an observable route to otherwise abstract topological data. *Caveat:* localizes and operationalizes the global datum; does not compute the character variety.

6. The curvature lever (Section 4). Observable low-order curvature constrains hidden higher-order/null structure via the Gauss equation—a cheap measurement of an expensive quantity.

6 The honesty ledger: what was retired

The credibility of the surviving core rests on the removal of what could not be supported. The following were retired, with reasons.

Retired 6.1 (Physical over-claims). Kolmogorov turbulence recovery (circular, and an arithmetic error $k \cdot k^{-4/3} = k^{-1/3} \neq k^{-5/3}$); a π-diagnosis; emergent spacetime; a Fisher-induced Lorentz signature read as gravity; Fisher-rank-loss "seepage"; quantum/biological anchorings; and a circular Omori-law validation (rewritten as a documented negative result). None had support; each was removed by the framework's own falsification discipline.

Remark 6.2 (Falsification-first). Every load-bearing theorem was numerically verified before being written, and retirements are treated as the method working, not as failures. The reactive Lorentzian signature (Semi-win 3.2) is retained precisely because it is stated as an exact congruence result and *not* as physics.

7 Scope, non-claims, and boundaries

Non-claim. The framework is a geometry of *observation*. Its objects—connection, curvature, holonomy, second fundamental form, Lorentzian signature, geometric phase—are those of statistics, estimation, and submanifold/bundle geometry. Their formal identity with objects in physical geometry (gauge fields, spacetime, Berry phase) is *resemblance*, and no physical theory, field dynamics, or spacetime is claimed. The global classification is *reduced and bounded*, not enumerated: computing the character variety for rich $\pi_1(B)$ is the standard hard moduli problem. The Fisher-insufficiency and efficiency results are stated for the Gaussian model; the bundle results assume fixed similarity type and a base with standard niceness hypotheses.

On significance. Internal coherence and correctness are established here; *significance*—whether the organizing lens is useful enough to be adopted—is a judgment for the field, requiring exposure to specialists in information geometry and statistical estimation. This paper makes the case; it does not pre-empt that judgment.

8 Conclusion

Resolution geometry is one object studied through one invariant, carried coherently from local structure to global classification to second-order geometry, with every load-bearing claim verified and every over-claim retired. Its core is classical mathematics resynthesized around the resolved–null coupling, with a few genuinely non-trivial theorems—the irreducibility of the coupling, the reactive Lorentzian signature—and one striking identity: estimation efficiency equals sector mutual information. Its reason to exist in practice is the curvature bridge: the coupling is statistical at first order and geometric at second, joined by the Gauss equation, so cheap observations lever hidden structure. What it offers is concrete and quantified; what it claims is bounded and honest; what it leaves open is delimited. The coupling was the thread throughout, and it holds the whole framework together.

A The machine on one example

On a single system ($r = q = 3$ blocks; $n = 4$ reactive form; a generic map $\mathbb{R}^2 \to \mathbb{R}^5$ for curvature) the layers run end to end:

- **Efficiency = information:** gain $= 0.136775 = I(y_R; y_N)$ (both sides equal).
- **Reactive signature:** $(4, 0)$ below the fundamental, $(3, 1)$ in the first window — Lorentzian $(1, n - 1)$.
- **Curvature bridge:** intrinsic curvature -0.51908 = second-fundamental-form curvature -0.51908 (Gauss); sphere control returns $1/R^2$ exactly.
- **Compact transport:** holonomy powers $\|A^k\|$ constant (1.732) — bounded, well-posed.

The single machine thus exhibits, on one object, the estimation win, the reactive semi-win, the curvature spine, and the global well-posedness.

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